Tiago Ferrão-Lindqvist

Data Engineering | Test Data Automation & Analytics

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PROFILE

Mechanical engineer with 12 years of experience in product design, testing and validation, now transitioning into data engineering. Builds Python tools to process test-rig data, apply specification-based quality checks and generate reports. Experience querying SQL Server test data, writing summaries to a staging table and building Power BI dashboards used by engineering teams. Currently completing an online data engineering course.

DATA ENGINEERING PROJECTS

Rig Data Processing and Reporting Tool | Corvane Automotive Components *2021–Present*

- Developed Python scripts to parse TDMS and CSV output from six test rigs, processing approximately 2 GB of data per week.
- Added checks that flag channels outside test specifications and generate standard report figures; reduced report preparation from about three days to half a day per test.
- Scheduled nightly runs on a lab PC using Windows Task Scheduler. The tool writes summary results to a staging table in the plant's SQL Server database; IT created the table.
- Packaged the tool for internal use; four other test engineers now use it for their tests.
- Technologies: Python, pandas, NumPy, Matplotlib, npTDMS, SQL Server.

Test Results Dashboard | Corvane Automotive Components *2023*

- Built a Power BI dashboard using the staging table to show pass/fail trends and cycles to failure across product generations.
- Dashboard used by approximately 15 engineers in validation and design.
- Technologies: Power BI, SQL.

Bike-Sharing Trips API | Personal Learning Project *2025*

- Built a small Flask API over a public bike-sharing dataset loaded into SQLite, with three read-only endpoints; runs locally on a laptop.
- Followed a tutorial to run the project with Docker Compose.
- Technologies: Python, Flask, SQLite, Docker Compose, Git.

PROFESSIONAL EXPERIENCE

Test & Validation Engineer | Corvane Automotive Components, Setúbal *Mar 2019–Present*

- Plan and run durability, vibration and climatic tests on seat-adjuster mechanisms across six servo-hydraulic and electric test rigs; prepare reports for OEM customers.
- Correlate rig results with FEA using ANSYS Mechanical and lead 8D root-cause analysis on field failures.
- Proposed a redesigned adjuster rail bracket that reduced warranty returns for the part by 40%.
- Query the plant's SQL Server test-results database for trend analysis across three product generations.
- Maintain MATLAB/Simulink models of the adjuster drive used to select rig load profiles.
- Trained three apprentices in rig safety and test setup (2022–2024).

Mechanical Design Engineer | Halvik Pumps, Porto *Sep 2014–Feb 2019*

- Designed centrifugal pump casings and impellers in SolidWorks, prepared production drawings with GD&T and owned DFMEA for two product lines.
- Performed structural FEA of casings in ANSYS; supported a material change from grey cast iron to ductile iron that reduced material cost by 12% for one casing family.
- Wrote a LabVIEW data-acquisition program for the pump performance test bench, reducing test setup time from about two hours to 30 minutes.
- Supported conversion of the performance test bench to a PLC-controlled setup, changing setpoints and timers in TIA Portal with an automation engineer.

EDUCATION

MSc Mechanical Engineering | Sado Valley Polytechnic *2014*

Thesis: Fatigue life prediction of welded brackets under variable-amplitude loading

PROFESSIONAL DEVELOPMENT

Online Data Engineering Course | In progress; 6 of 9 modules completed

TECHNICAL SKILLS

Programming and data: Python (pandas, NumPy, Matplotlib, npTDMS), SQL, MATLAB/Simulink, Excel/VBA
Data and reporting: Power BI, Minitab, LabVIEW
Engineering: ANSYS Mechanical, SolidWorks, CATIA V5, design of experiments (DOE), GD&T, DFMEA, 8D problem solving
Personal project tools: Flask, SQLite, Docker Compose, Git

LANGUAGES

Portuguese (native) | English (fluent) | Spanish (good)